

Lecture 5:

AI504 - Knowledge Representation

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Contents

Recall	3
Terminology	3
Prooftrees	3
Soundness	3

Recall

Recall the logic of “all p are q ”

Now fix an underlying set P

- a P -sentence is just a phrase “all $\underbrace{p \text{ are } q}_{(p,q)\text{-mathematically}}$, where $p, q \in P$ ”
- a P -model is a set M and a function $P \rightarrow \mathcal{P}(M)$, if $p \in P$, then $\llbracket p \rrbracket \subset P$

Terminology

P is the signature of a P -model or P -sentence

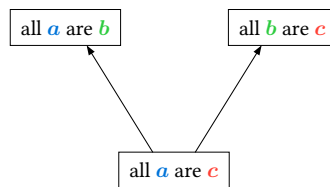
Suppose $\mathcal{M}$ is a P -model and “all p are q ” is P -sentence

Then $\mathcal{M} \models \text{all } p \text{ are } q$ means $\llbracket p \rrbracket \subset \llbracket q \rrbracket$

Suppose Γ is a subset of sentences and ϕ is a single sentences, then $\Gamma \models \phi$ mean that for all models $\mathcal{M}$, if $\mathcal{M} \models \Gamma$, then $\mathcal{M} \models \phi$

Prooftrees

A proof tree is a binary tree labeled by sentences such that at each non-leaf node



Suppose that T is a proof tree. Then T is a **proof** of ϕ from Γ if:

1. ϕ labels the root of T
2. every leaf of T either is in Γ or has the form $\underbrace{\text{all } _ \text{ are } _}_{\text{blanks are equal}}$

Definition: $\Gamma \vdash \phi$

$\Gamma \vdash \phi$ if there is a proof tree of ϕ from Γ

During this and the next lecture we want to prove

$$\Gamma \models \phi \iff \Gamma \vdash \phi$$

where

$$\underbrace{\Gamma \models \phi \implies \Gamma \vdash \phi}_{\text{Soundness } (\models \text{ implies } \vdash)}$$

and

$$\underbrace{\Gamma \vdash \phi \implies \Gamma \models \phi}_{\text{Completeness } (\vdash \text{ implies } \models)}$$

Soundness

Definition: Soundness

For any set of sentences Γ and sentence ϕ , if $\Gamma \vdash \phi$ then $\Gamma \models \phi$

For all Γ, ϕ : if there exists a proof of ϕ from Γ , then for every model $\mathcal{M}$, if $\mathcal{M} \models \Gamma$ then $\mathcal{M} \models \phi$.

Message

for all proofs T , Γ and ϕ .

If T is a proof of ϕ from Γ then for every model $\mathcal{M}$,

if $\mathcal{M} \models \Gamma$ then $\mathcal{M} \models \phi$

Now we can do induction to prove this, where goal

if T is a proof of ϕ from Γ and $\mathcal{M}$ is any model satisfying Γ then $\mathcal{M} \models \phi$ too

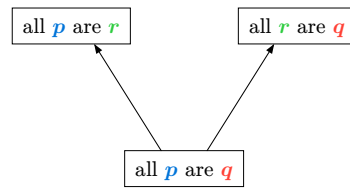
Proof by induction

Base Case) (Assume proof T is a single leaf)

- 1 | Assume proof T is a single leaf
- 2 | Then ϕ is the only thing in T , since T is a proof from ϕ .
- 3 | Notice that all the leaves from T fomrs from Γ
- 4 | Let $\mathcal{M}$ be any model that satisfies (all sentences in) Γ
- 5 | since $\phi \in \Gamma$, $\mathcal{M} \models \phi$

Inductive step

- 6 | Suppose that T is not a leaf
- 7 | Decompose T as follows. Say that ϕ is “all p are q ”
- 8 | Then there exists an r s.t T looks like



then let T_0, T_1 be the left & right halves of T

T_0 is a proof of “all p are r ” from Γ (**IH** but explain better)

T_1 is a proof of “all r are q ” from Γ (**IH** but explain better)

Suppose that $\mathcal{M} \models \Gamma$.

By **IH** applied to T_0 , $\mathcal{M} \models$ all p are r

By **IH** applied to T_1 , $\mathcal{M} \models$ all r are q

Since $\mathcal{M} \models$ all p are r

$\llbracket p \rrbracket \subset \llbracket r \rrbracket$

Since $\mathcal{M} \models$ all r are q

$\llbracket r \rrbracket \subset \llbracket q \rrbracket$

Therefore,

$\llbracket p \rrbracket \subset \llbracket q \rrbracket$, so $\mathcal{M} \models$ all p are q